WORKED SOLUTION

STEP II 1989, Question 5

Pure • Polar coordinates

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s2-q5

The key idea

Part (i) is just $\mathrm{d}y/\mathrm{d}x = (\mathrm{d}y/\mathrm{d}\theta)/(\mathrm{d}x/\mathrm{d}\theta)$ with $x = r\cos\theta$, $y = r\sin\theta$. For the mirror, polar coordinates centred on O are the natural choice, because the reflected ray always runs along the radius back to the pole. The law of reflection then tells us the direction of the normal at every point in terms of θ alone, and part (i) turns that into a first-order differential equation for $r(\theta)$ that separates.

Solution

Part (i)

A point with polar coordinates (r, θ) has Cartesian coordinates $x = r\cos\theta$, $y = r\sin\theta$. Along a curve $r = r(\theta)$, both are functions of θ, and by the product rule

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta, \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta.$$

By the chain rule (for a parametrised curve), wherever $\mathrm{d}x/\mathrm{d}\theta \neq 0$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}\theta}{\mathrm{d}x/\mathrm{d}\theta} = \frac{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\sin\theta + r\cos\theta}{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\cos\theta - r\sin\theta} = \frac{\dfrac{\mathrm{d}r}{\mathrm{d}\theta}\tan\theta + r}{\dfrac{\mathrm{d}r}{\mathrm{d}\theta} - r\tan\theta},$$

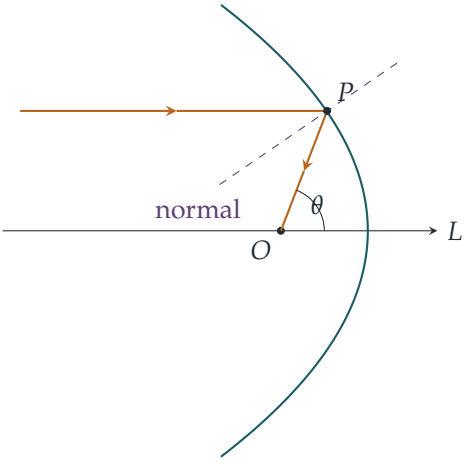
dividing top and bottom by $\cos\theta$, which is allowed provided $\cos\theta \neq 0$. This is the required formula.

Part (ii)

Reducing to a plane curve. Take any plane Π containing L, and a point P of the mirror in Π. The ray through P parallel to L lies in Π, and so does O; hence the reflected ray PO lies in Π too. The normal at P bisects the angle between two lines in Π, so it also lies in Π. The curve C in which the mirror meets Π has its tangent at P in both Π and

the mirror's tangent plane, so it is perpendicular to the normal. So, within the plane Π, the normal to C at P bisects the angle between the incident and reflected rays: we have a two-dimensional problem.

Coordinates. In Π use polar coordinates with pole O and initial line along L, oriented so that the incoming rays travel in the direction $\theta = 0$ (as in the figure, they come from the side of L beyond the mirror's opening). Let $P = (r, \theta)$ be on C.



The direction of the normal. From P, the incident ray goes back in the direction $(-1, 0)$ and the reflected ray goes towards O in the direction $(-\cos\theta, -\sin\theta)$. Both are unit vectors, so their sum points along the bisector of the angle between them, which is the normal:

$$(-1 - \cos\theta,\ -\sin\theta) = -2\cos\tfrac{\theta}{2}\,(\cos\tfrac{\theta}{2},\ \sin\tfrac{\theta}{2}),$$

using $1 + \cos\theta = 2\cos^2\frac{\theta}{2}$ and $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$. So the normal at P makes angle $\frac{\theta}{2}$ with L, the tangent is perpendicular to it, and the gradient of C at P is

$$m = -\cot\tfrac{\theta}{2}.$$

The differential equation. Write $r' = \mathrm{d}r/\mathrm{d}\theta$, $t = \tan\theta$ and $\tau = \tan\frac{\theta}{2}$, so that $t = \dfrac{2\tau}{1-\tau^2}$ and $m = -\dfrac{1}{\tau}$. By part (i),

$$\frac{r't + r}{r' - rt} = m \quad\Longrightarrow\quad r'(t-m) = -r(1+mt) \quad\Longrightarrow\quad \frac{r'}{r} = -\frac{1+mt}{t-m}.$$

Now

$$1 + mt = 1 - \frac{2}{1-\tau^2} = -\frac{1+\tau^2}{1-\tau^2}, \qquad t - m = \frac{2\tau}{1-\tau^2} + \frac{1}{\tau} = \frac{1+\tau^2}{\tau(1-\tau^2)},$$

and so

$$\frac{1}{r}\frac{\mathrm{d}r}{\mathrm{d}\theta} = \tan\tfrac{\theta}{2}.$$

Ask yourself Part (i) breaks down where $\cos\theta = 0$. Does that leave a gap?

No. The equation $r'/r = \tan\frac{\theta}{2}$ involves only continuous functions of θ on $-\pi < \theta < \pi$. It holds at every point except possibly the isolated values $\theta = \pm\frac{\pi}{2}$ (and any isolated points where the tangent is vertical), so by continuity of r' it holds there too. (Alternatively, the angle ψ between the radius and the tangent satisfies $\tan\psi = r/r'$, and the geometry gives $\psi = \frac{\pi}{2} - \frac{\theta}{2}$ directly, with no exceptional points.)

Solving. Integrating, with $\cos\frac{\theta}{2} > 0$ for $-\pi < \theta < \pi$,

$$\ln r = \int \tan\tfrac{\theta}{2}\,\mathrm{d}\theta = -2\ln\cos\tfrac{\theta}{2} + \text{const}, \qquad \text{so} \qquad r = \frac{k}{\cos^2\frac{\theta}{2}} = \frac{2k}{1+\cos\theta}$$

for a constant $k > 0$ (as $r > 0$). Put $\ell = 2k$. Then $r + r\cos\theta = \ell$, that is $r = \ell - x$. Squaring, using $r^2 = x^2 + y^2$,

$$x^2 + y^2 = (\ell - x)^2 \quad\Longleftrightarrow\quad y^2 = \ell^2 - 2\ell x.$$

This is a parabola with axis L and focus O (it is $Y^2 = 4aX$ with $X = \frac{\ell}{2} - x$, $a = \frac{\ell}{2}$, so the focus is at $x = \frac{\ell}{2} - a = 0$). Since Π was any plane containing L, the mirror meets every such plane in a parabola, as required. (In fact the constant ℓ is the same in each plane only if the mirror is a single paraboloid of revolution; the question does not need this.)

Checks

- At $\theta = 0$ we get $r = \frac{\ell}{2}$: the vertex is halfway between O and the directrix $x = \ell$, as it should be for a parabola with focus O.
- For $\ell = 2$ and $\theta = \frac{\pi}{2}$ the point is $P = (0, 2)$, the gradient of $y^2 = 4 - 4x$ there is $-1 = -\cot\frac{\pi}{4}$, and reflecting the direction $(1, 0)$ in this tangent gives $(0, -1)$, straight down to O.
- Numerically: the gradient formula in (i) agreed with finite differences on 2000 random curves (to 10^{-8}); a Runge–Kutta solution of $r' = r\tan\frac{\theta}{2}$ from $r(0) = 1.35$ gave $r(2.5) = 13.5776380$, matching $2.7/(1 + \cos 2.5)$; and reflecting a ray parallel to L off $y^2 = \ell^2 - 2\ell x$ sent it through O at 1000 random points.

Common slips

- Taking the wrong bisector: the sum of the two unit vectors pointing *away* from P along the rays gives the normal; their difference gives the tangent.
- Forgetting to explain why the problem is planar: the plane through L contains the incident ray, O, hence the reflected ray and the normal.
- Writing $\ln r = -2\ln\cos\frac{\theta}{2}$ without a constant, which loses the whole family of parabolas.

Beyond STEP

The equation $r(1+\cos\theta) = \ell$ is the polar form of a conic with focus at the pole and eccentricity 1; replacing 1 by e gives ellipses and hyperbolas, which is exactly how planetary orbits appear when one solves Newton's equations in polar coordinates. The reflection argument is also the reason satellite dishes and car headlamps are paraboloids. Kleppner and Kolenkow, *An Introduction to Mechanics*, derives the conic orbits this way.

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